5. Market Heterogeneity and Nonlinear Effects of the Expulsion Mechanism

The preceding analyses examine the baseline effect of the expulsion mechanism, market survival and collapse risk, and outcomes under alternative expulsion thresholds. Together, these analyses assess the overall role of expulsion, its implications for market survival, and the consequences of changing the punishment rule. They leave open an important question: even under the same expulsion rule, does its effect vary across markets with different initial conditions? For example, markets with a higher initial share of cheaters, or with honest traders who are more susceptible to becoming cheaters, may not obtain the same survival gains from expulsion.

This section addresses this question by examining heterogeneity in the effect of a fixed expulsion rule. Holding the expulsion threshold at $q = 3$, we vary the initial share of cheaters, ρ, and the threshold governing the conversion of honest traders, p. We then estimate the conditional average treatment effect (CATE) of expulsion on market survival time. Because the relationship between these market characteristics and the policy effect may be nonlinear, we compare random forests, gradient boosting, and an R-learner with conventional regression benchmarks. Independent test samples are used to assess whether these methods can predict policy effects in simulated markets not observed during training.

5.1 Simulation Design and Learning Target

To ensure comparability with Tasks A–C, we retain the behavioral rules, common parameters, and definition of market survival from the baseline simulation in Task A. We vary only two market characteristics: the initial share of cheaters (ρ), which measures the initial prevalence of fraudulent behavior, and the conversion threshold for honest traders (p), which governs how readily honest traders become cheaters.

Specifically, ρ ranges from 0.05 to 0.40 in increments of 0.01, yielding 36 values, while p takes values in $\{2, 3, 4\}$. Their combinations produce 108 market configurations. Each configuration is simulated under both expulsion states using 80 random seeds, resulting in 17,280 simulation runs (8,640 paired experiments). Other parameters follow the baseline: $N = 100$ agents, $a = 3$, expulsion threshold $q = 3$, initial wealth of 10, natural consumption $c = 0$, and an observation horizon $T = 200$. The bankruptcy mechanism is enabled throughout ($B = 1$). A market is classified as collapsed when the number of active honest traders first falls to 10% or less of its initial level. The outcome is restricted survival time, $Y = \min(T_{collapse}, 200)$, so $Y = 200$ if the market does not collapse during the observation period. This is the same outcome as rmst_T in the preceding analysis.

Let $D = 1$ denote a market with expulsion and $D = 0$ a market without expulsion. Our estimand is the conditional average treatment effect, $\tau(\rho, p) = E[Y(1) - Y(0) \mid \rho, p]$, where $Y(1)$ and $Y(0)$ denote restricted survival time under the two regimes. The effect is measured in simulation steps; $\tau > 0$ indicates that expulsion increases expected survival time. For each configuration and random seed s, we simulate both regimes and construct the paired difference $\Delta_s = Y_s(1) - Y_s(0)$. Using the same seed for the two regimes reduces noise in their comparison. The 800 runs that overlap with the baseline experiment reproduce the earlier restricted survival outcomes exactly, confirming consistency with the common setup. Importantly, treatment contrasts are generated directly by controlled counterfactual simulations. Machine learning is used to estimate and predict how $\tau(\rho, p)$ varies across market conditions, rather than to identify a causal effect from observational data.

5.2 Estimation Methods

The paired simulations yield a direct measure of the policy effect for each configuration and seed, turning the machine learning problem into one of learning a heterogeneous-effect function. Let $X = (\rho, p)$ denote the market characteristics. Random forests (RF; Breiman, 2001) and gradient boosting (GB; Friedman, 2001) directly regress the paired differences Δ_s on X. We also implement the R-learner of Nie and Wager (2021) as an alternative approach to heterogeneous-effect estimation. It uses individual simulation outcomes from both treatment states. Within the training sample, we first estimate $m(X) = E[Y \mid X]$ using four-fold cross-fitting with folds grouped by random seed. Given the known treatment propensity $P(D = 1 \mid X) = 0.5$, we then form the pseudo-outcome $Z = (Y - \hat{m}(X))/(D - 0.5)$ and fit it as a function of X using gradient boosting. Because $(D - 0.5)^2 = 0.25$ for every observation, the corresponding weighted squared-error loss is equivalent to ordinary least squares on this pseudo-outcome. All nuisance-function estimation is confined to the training sample to avoid test-data leakage. To assess the added predictive value of machine learning, we compare these methods with linear interaction, quadratic, and piecewise regression models. Each

regression benchmark directly models the paired differences; the breakpoint in the piecewise specification is selected using validation data.

5.3 Out-of-Sample Predictive Performance

Having specified the estimators, we assess their ability to predict policy effects outside the training sample. We hold out observations along two dimensions: random seeds and market configurations. Seeds 1–48 are used for training, 49–64 for validation and model selection, and 65–80 exclusively for final testing. In addition, all configurations with ρ = 0.08, 0.13, 0.18, 0.23, 0.28, 0.33, or 0.38 are excluded from training and validation. We thereby construct two tests. The "new seeds" test uses new random seeds for the remaining 29 values of ρ that appeared during training. The "held-out configurations" test uses new seeds for the seven values of ρ absent from training. Both tests include p = 2, 3, and 4. All methods use the same split. After hyperparameters have been selected, each model is refitted using seeds 1–64 for the 29 training values of ρ.

Our main evaluation metrics are calculated at the configuration level. For each configuration, the paired differences are averaged over the 16 independent test seeds to obtain a test-sample estimate of the CATE. Predicted effects are then compared with these configuration-level averages to calculate the mean absolute error (MAE) and root mean squared error (RMSE), giving equal weight to each configuration. We also report pair-level RMSE, which includes run-specific simulation noise and is therefore typically larger. Finally, holding the fitted models fixed, we resample the 16 test seeds 300 times and report the bootstrap standard deviation of the test error. This quantity captures variation due to the finite test sample but excludes training and hyperparameter-selection uncertainty; it should not be interpreted as a confidence interval for population prediction error.

Table 3. Out-of-Sample Prediction Errors for the Effect of Expulsion

Test sample	Method	Config. MAE	Config. RMSE	Pair RMSE	Boot. SD
New seeds	Linear interact.	40.64	46.85	54.93	0.69
New seeds	Quadratic	40.62	46.83	54.91	0.69
New seeds	Piecewise	32.04	38.91	48.33	0.78
New seeds	RF	1.98	4.58	29.03	0.63
New seeds	GB	2.06	4.66	29.04	0.62
New seeds	R-learner	2.15	4.74	29.05	0.63
Held-out ρ	Linear interact.	36.84	43.72	54.17	1.28
Held-out ρ	Quadratic	37.07	43.67	54.13	1.30
Held-out ρ	Piecewise	31.82	39.50	50.83	1.75
Held-out ρ	RF	6.06	15.49	35.55	1.99
Held-out ρ	GB	4.71	12.77	34.45	1.57
Held-out ρ	R-learner	5.46	14.97	35.32	1.61

Note: B = 1 and q = 3. All errors are measured in simulation steps. "New seeds" comprises 87 configurations and 1,392 paired test observations; "held-out configurations" comprises 21 configurations and 336 paired test observations, with "held out" referring only to values of ρ not seen during training. RF denotes random forests; GB denotes gradient boosting. Piecewise regression uses a common breakpoint in ρ across all values of p and includes interactions with p. Model settings are selected by configuration-level MAE on the validation sample. Bootstrap SD is the bootstrap standard deviation of test error.

Table 3 shows that all three machine learning methods outperform the regression benchmarks in both out-of-sample tests. For new seeds, RF achieves the lowest configuration-level MAE (1.98 steps), followed by GB (2.06) and the R-learner (2.15), compared with 40.64 for linear interaction regression. In the more demanding test on held-out configurations, GB obtains an MAE of 4.71 steps, compared with 6.06 for RF, 5.46 for the R-learner, and 31.82 for piecewise regression. Within the parameter range considered here, these comparisons indicate that machine learning methods can more accurately predict the nonlinear heterogeneity in expulsion effects. GB is selected as the principal predictive model based on validation performance, independently of the final test sample. Having established predictive accuracy, we next examine the shape of the estimated effect function and its implications for different market conditions.

5.4 Nonlinear Effects and the Effective Region

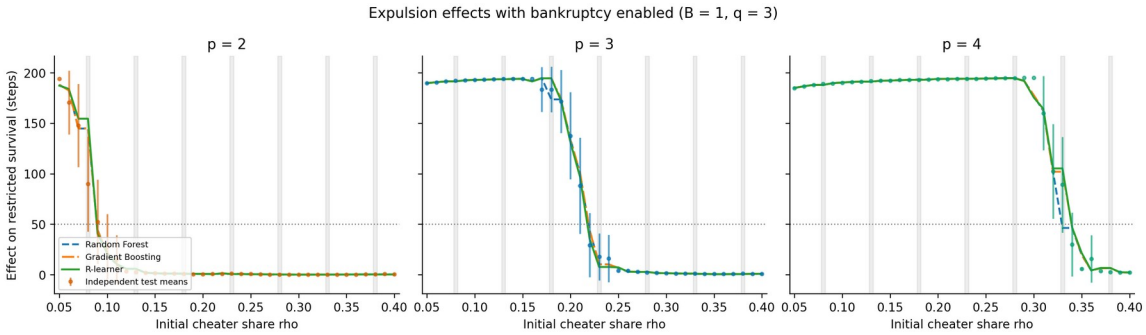


Figure 3. Predicted effects of expulsion across market conditions (B = 1, q = 3). Curves are fitted using the 29 training values of ρ and seeds 1–64; points show mean paired effects from the independent test sample for all 36 values of ρ, with approximate pointwise 95% intervals. Gray vertical bands mark values of ρ excluded from training. The horizontal dashed line marks a reference gain of 50 survival steps. RF, GB, and R-learner denote the three machine learning estimators.

To interpret the heterogeneity learned by the models, Figure 3 plots the effect of expulsion against the initial share of cheaters, separately for each conversion threshold p. The effect does not change smoothly with ρ: expulsion substantially prolongs survival when the initial share of cheaters is low, but its benefit declines sharply once ρ reaches a certain range. Moreover, increasing p shifts the decline to higher values of ρ. Within the simulated environment, this pattern is consistent with honest traders being less likely to convert into cheaters, allowing expulsion to remain effective even when the initial share of cheaters is relatively high. For descriptive purposes, we call configurations with predicted survival gains exceeding 50 steps the "effective region." The 50-step cutoff is a reference threshold, not a statistically identified structural breakpoint or a welfare-optimal policy rule. The prediction errors in Table 3 do not depend on this cutoff.

Table 4. Predicted Boundaries of the Effective Region by Conversion Threshold

p	GB crossing interval	Test boundary	Bootstrap range
2	0.08—0.09	0.09	0.07—0.10
3	0.21—0.22	0.21	0.21—0.22
4	0.33—0.34	0.33	0.32—0.34

Note: The model crossing interval consists of the adjacent grid points, spaced 0.01 apart in ρ, between which the GB-predicted effect falls from above to below 50 steps. The test boundary point is the last grid point before the independent test-sample mean effect first drops below 50 steps. The bootstrap range is obtained by resampling test seeds while holding the model fixed; it measures variation in the boundary on the discrete grid, excludes estimation uncertainty, and is not a confidence interval for a continuous threshold.

Table 4 shows that the GB-predicted crossing intervals are $\rho \in [0.08, 0.09]$, $[0.21, 0.22]$, and $[0.33, 0.34]$ for p = 2, 3, and 4, respectively. The independent test estimates display a similar rightward shift, although the boundary points do not coincide exactly. We also calculated sensitivity results for 25-step and 100-step cutoffs. Thus, the 50-step criterion should not be regarded as a unique economic threshold, and the effective region is defined only on the current discrete parameter grid. Under the simulation rules, a higher p delays conversion of honest traders and may leave more time to report and expel cheaters. This explanation is consistent with the model's mechanics, but the predicted effect curves do not constitute a causal mediation decomposition of that channel.

5.5 Feature Importance

Table 5. Permutation Importance of Market Characteristics

Feature	MAE increase	Permutation SD
ρ	59.18	16.26
p	56.71	15.07

Note: On the held-out configuration test sample, we independently permute ρ or p 100 times and record the increase in the configuration-level MAE of the selected GB model relative to its unpermuted value. The permutation standard deviation reflects variation across random permutations, not a sampling confidence interval.

In addition to examining the estimated effect curves, we use permutation importance to assess the predictive role of the two market characteristics. As Table 5 shows, permuting ρ increases configuration-level MAE by 59.18 steps on average, while permuting p increases it by 56.71 steps. Both features therefore contain substantial information for predicting the effect of expulsion. Permutation importance measures the deterioration in predictive accuracy when a feature is shuffled, and depends on the parameter distribution, the test sample, and the estimation method. It should not be interpreted as a causal contribution or as a percentage decomposition of outcome variation. Because ρ and p form a complete factorial design, permuted combinations remain within the parameter domain studied here.

5.6 Scope and Limitations

Three qualifications are important. First, configuration-level test averages are themselves noisy estimates of the CATE, especially near regions where the effect changes rapidly. The final test uses only 16 random seeds, and the bootstrap results do not incorporate uncertainty from model training or hyperparameter selection. Second, all held-out values of ρ lie within the original parameter interval. The held-out configuration test therefore evaluates interpolation over $\rho \in [0.05, 0.40]$ and $p \in \{2, 3, 4\}$, rather than extrapolation beyond these values. Third, the simulation makes the initial share of cheaters directly observable, whereas this quantity may be difficult to measure in real trading platforms. This limits the immediate practical applicability of the predictions.

Furthermore, q is held fixed at 3, so this analysis does not identify an optimal expulsion threshold. A full welfare assessment would also need to account for reporting and enforcement costs and net gains from trade. In sum, Tasks A–C focus on baseline policy effects, market survival, and changes in the expulsion threshold. Task D complements those analyses by showing that, under a given expulsion rule, the effect on market survival depends nonlinearly on market characteristics. With training, validation, and test observations kept separate, the machine learning methods attain lower out-of-sample prediction errors than the selected regression benchmarks, providing additional evidence on the heterogeneous performance of the mechanism across simulated markets.

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